Constrained Optimization

Constrained optimization

1

Minimize $f(\boldsymbol{x})$

subject to:

$$g_i(x) < 0, \quad i = 1,2,..,m$$
$$h_j(x) = 0, \quad j = 1,2,..,l$$

Where $x \in \Omega \subseteq \mathbb{R}^n$ and f, g_i, h_i usually assumed to be in C^2 (i.e., with continuous second partial derivatives)

- What, exactly, is the definition of a minimum? local and global
- Does a solution even exist? feasibility
- Is it unique? if it's convex, then yes
- What are the necessary & sufficient conditions to be a solution?
- How do we solve?

Equality constraints: Lagrange multiplier

Minimize $f(\boldsymbol{x})$

subject to:

$$h_j(x) = 0, \quad j = 1,2,..,l$$

Where $x \in \Omega \subseteq \mathbb{R}^n$ and f, h_i usually assumed to be in C^2 (i.e., with continuous second partial derivatives)

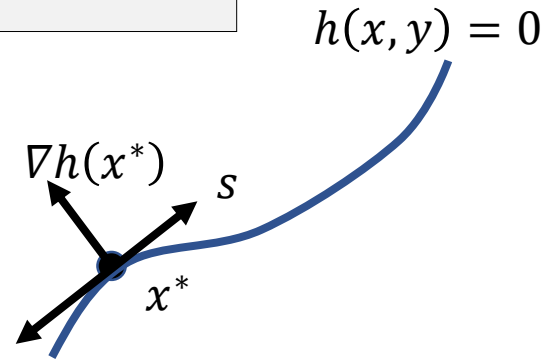
Aim: Find optimal x^* of f while satisfying the constraints $h_j(x)$.

2D problem: Minimize $f(\boldsymbol{x}, \boldsymbol{y})$

subject to:

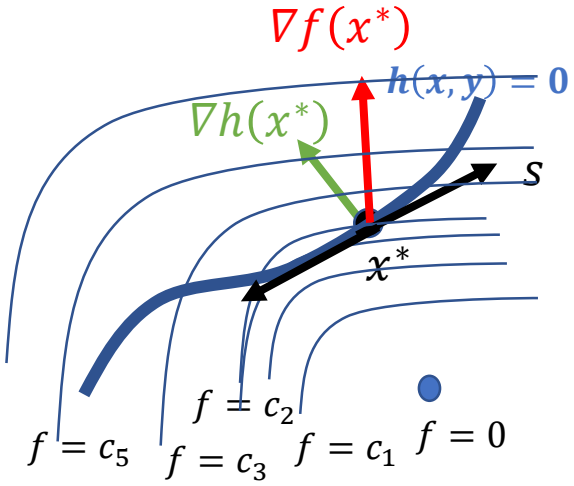
$$h(x, y) = 0,$$

Aim: Find optimal $x^* = (x^*, y^*)$ of f while satisfying the constraints $h(x, y) = 0$.



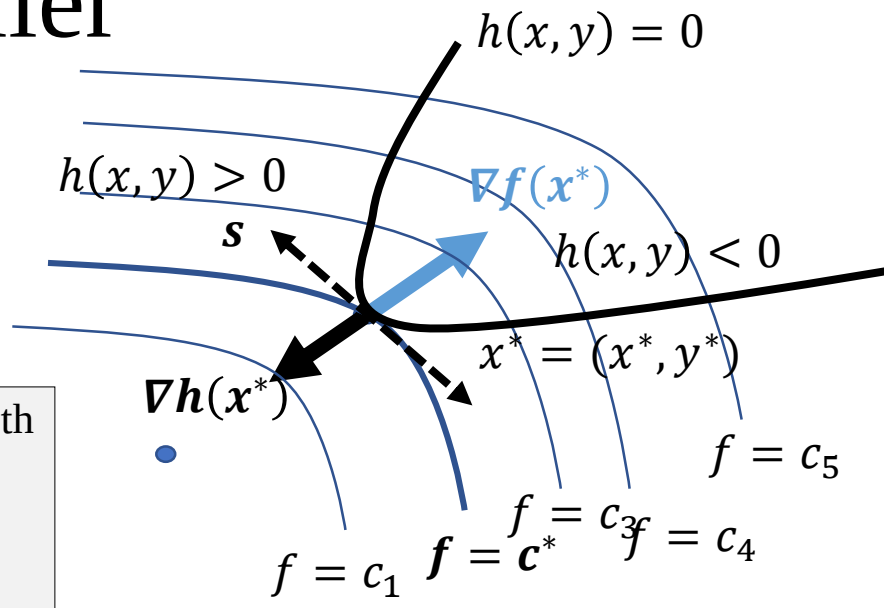
At a point x^*that lie on the constraint surface $h(x, y) = 0$,

- **Feasible direction:**
  - A feasible direction s is any direction that keeps the point on the surface after a small step. It means:
    - $s^T \nabla h(x^*) = 0$ ensuring $h(x^* + \alpha s) = 0$ for small α.
- **Descent direction**
  - A direction s is a descent direction if $s^T \nabla f(x^*) < 0$ (f is decreasing in s direction).



Equality constraints: Lagrange multiplier

- At an extreme point (a local minimum or maximum on the constraint surface), the gradients $\nabla f(x^*)$ and $\nabla h(x^*)$ are parallel, so no feasible direction exists that further decreases the objective function, satisfying the optimality condition. . Or no feasible direction can be a descent direction

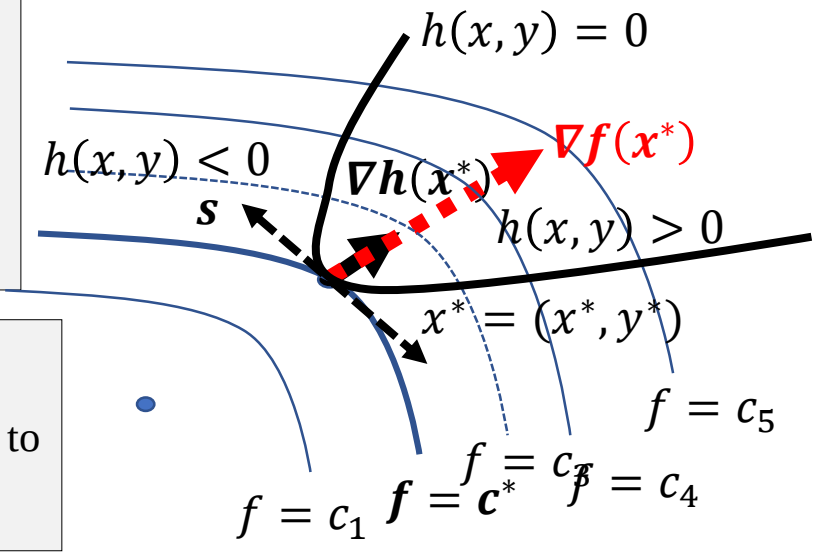


Case: when $\lambda > 0$

- At a **candidate point (or stationary point)** x^*, the feasible direction s is orthogonal to both gradients ∇f and ∇h. This means that ∇f and ∇h must be **collinear.**
- Phrased differently, there exists some $\lambda \in \mathbb{R}$ such that

$$\nabla f(x^*) + \lambda \nabla h(x^*) = 0$$

- $\lambda \in \mathbb{R}$ is a scalar (the **Lagrange multiplier**). This equation (Lagrange condition) indicates that $\nabla f(x^*)$ is balanced by a multiple (λ) of $\nabla h(x^*)$ at the same point.
  - **Large λ:** Large λ values imply the objective function is highly sensitive to the constraint, meaning the solution depends heavily on satisfying the constraint accurately.
  - **low λ:** it indicates that the constraint is either redundant or does not significantly impact the optimal solution.
- This **candidate point (or stationary point or Lagrange point)** x^*satisfies the Lagrange condition could be a min, max, or **saddle**.
- Extreme points: Stationary point + **verification (by using curvature (Hessian) or comparing values with plot helps)**
- **Stationary point may be** a min, max, or **saddle**
- Extreme points: min or max only



Case: when $\lambda < 0$

- Minimize (or maximize) an objective function $f(x)$ subject to a constraint $h(x) = 0$.
- Lagrange multipliers incorporate the **equability** constraint(s) into the optimization problem to convert it into **unconstraint optimization problem**

Equality constraints: Lagrange multiplier

- Define a **Lagrangian** function as:

$$L(x, y, \lambda) = f(x, y) + \lambda h(x, y)$$

  - Now $\nabla L = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda \frac{\partial h}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda \frac{\partial h}{\partial y} \\ h \end{pmatrix}$
  - Setting $\nabla L = 0 \Longrightarrow \begin{cases} \frac{\partial f}{\partial x} + \lambda \frac{\partial h}{\partial x} = 0 \\ \frac{\partial f}{\partial y} + \lambda \frac{\partial h}{\partial y} = 0 \\ h = 0 \end{cases}$

- From first two equations can be re-written as:
  - $\frac{\partial f}{\partial x} + \lambda \frac{\partial h}{\partial x} = 0 \Longrightarrow \frac{\partial f}{\partial x} = -\lambda \frac{\partial h}{\partial x}$
  - $\frac{\partial f}{\partial y} + \lambda \frac{\partial h}{\partial y} = 0 \Longrightarrow \frac{\partial f}{\partial y} = -\lambda \frac{\partial h}{\partial y}$
- Or, $\begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = -\lambda \begin{pmatrix} \frac{\partial h}{\partial x} \\ \frac{\partial h}{\partial y} \end{pmatrix}$
- $\Longrightarrow \nabla f(x, y) = -\lambda \nabla h(x, y)$ (Stationarity condition)

- **Stationarity condition:**
- The stationarity condition indicates that at a candidate point, the "force" from the gradient of the objective function is exactly balanced by the "forces" (represented by the gradient of the constraints, scaled by the multipliers) trying to keep you within the feasible region defined by the constraints.

- The value λ is known as the **Lagrange multiplier**. The approach of constructing the Lagrangian and setting its gradient to zero is known as the method of Lagrange multipliers.
- Instead of minimizing the Lagrangian directly, we seek the stationary points (x, y, λ) of L by setting its gradient to zero.

Equality constraints: Lagrange multiplier

Find the stationary or candidate points of the $f(x,y) = xy$ subject to the constraint $h(x,y) = \left(\frac{x^2}{8} + \frac{y^2}{2} - 1\right) = 0$, using the method of Lagrange multipliers.

- Construct the Lagrangian function as:

$$L(x,y,\lambda) = f(x,y) + \lambda h(x,y) = xy + \lambda\left(\frac{x^2}{8} + \frac{y^2}{2} - 1\right)$$

- Calculate the gradient $\nabla L(x,y,\lambda) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda\frac{\partial h}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda\frac{\partial h}{\partial y} \\ h \end{pmatrix} = \begin{pmatrix} y + \lambda\frac{x}{4} \\ x + \lambda y \\ \frac{x^2}{8} + \frac{y^2}{2} - 1 \end{pmatrix}$

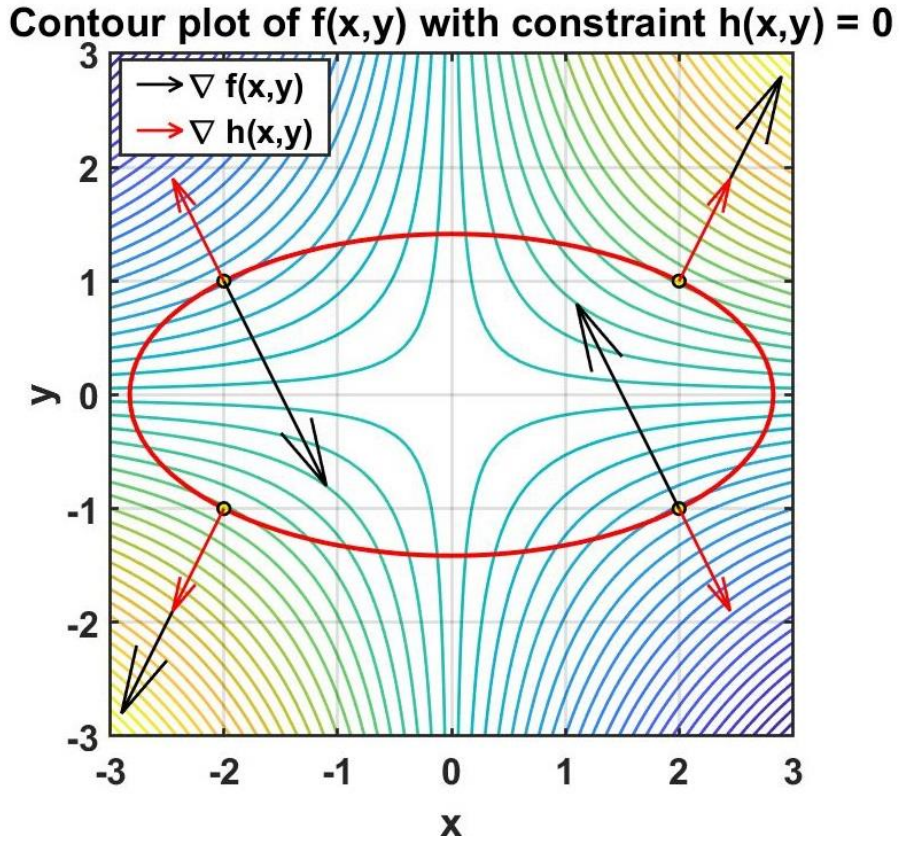
- Now $\nabla L(x,y,\lambda) = 0 \Rightarrow \begin{pmatrix} y + \lambda\frac{x}{4} \\ x + \lambda y \\ \frac{x^2}{8} + \frac{y^2}{2} - 1 \end{pmatrix} = 0 \Rightarrow \begin{cases} y + \lambda\frac{x}{4} = 0 & (1) \\ x + \lambda y = 0 & (2) \\ \frac{x^2}{8} + \frac{y^2}{2} - 1 = 0 & (3) \end{cases}$

- From (1) and (2), eliminating x we get: $y - \frac{\lambda^2}{4}y = 0 \Rightarrow y\left(1 - \frac{\lambda^2}{4}\right) = 0$
- Since $y \neq 0$ (otherwise $x = 0$ would hold by (2)), we get: $\lambda = \pm 2$
- Now from (2), $x = \pm 2y$, put this in (3) we get: $y = \pm 1$, and $x = \pm 2$

Equality constraints: Lagrange multiplier

- Stationary (or candidate or Lagrange) points : $(2,1), (-2,-1), (2,-1), (-2,1)$
- Function's maximum value is 2 achieved at: $(2,1)$ and $(-2,-1)$
- Function's minimum value is -2 achieved at: $(2,-1)$ and $(-2,1)$
- By manual verification and plot helps (elliptic curve as feasible space), all four are the extreme points (two maxima and two minima).

- $\nabla f(x,y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)^T = (y,x)^T$
  - $\nabla f(-2,1) = (1,-2)^T$
  - $\nabla f(2,-1) = (-1,2)^T$
- $\nabla h(x,y) = \left(\frac{\partial h}{\partial x}, \frac{\partial h}{\partial y}\right)^T = \left(\frac{x}{4}, y\right)^T$
  - $\nabla h(-2,1) = \left(-\frac{1}{2}, 1\right)^T$
  - $\nabla h(2,-1) = \left(\frac{1}{2}, -1\right)^T$
- At these points, $\nabla f(x,y)$ and $\nabla h(x,y)$ are opposite directions scaled by $\lambda = 2$.



Equality constraints: Lagrange multiplier

Find the Lagrange (stationary) points of the $f(x_1, x_2) = x_1 + x_2$ subject to the constraint $h(x, y)$: $(x_1^2 + x_2^2) = 2$. Additionally, determine the extreme points (maximum and minimum) of $f(x_1, x_2)$ without using second-order conditions, by manually verification.

- Construct the Lagrangian function as:

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda h(x_1, x_2) = x_1 + x_2 + \lambda(x_1^2 + x_2^2 - 2)$$

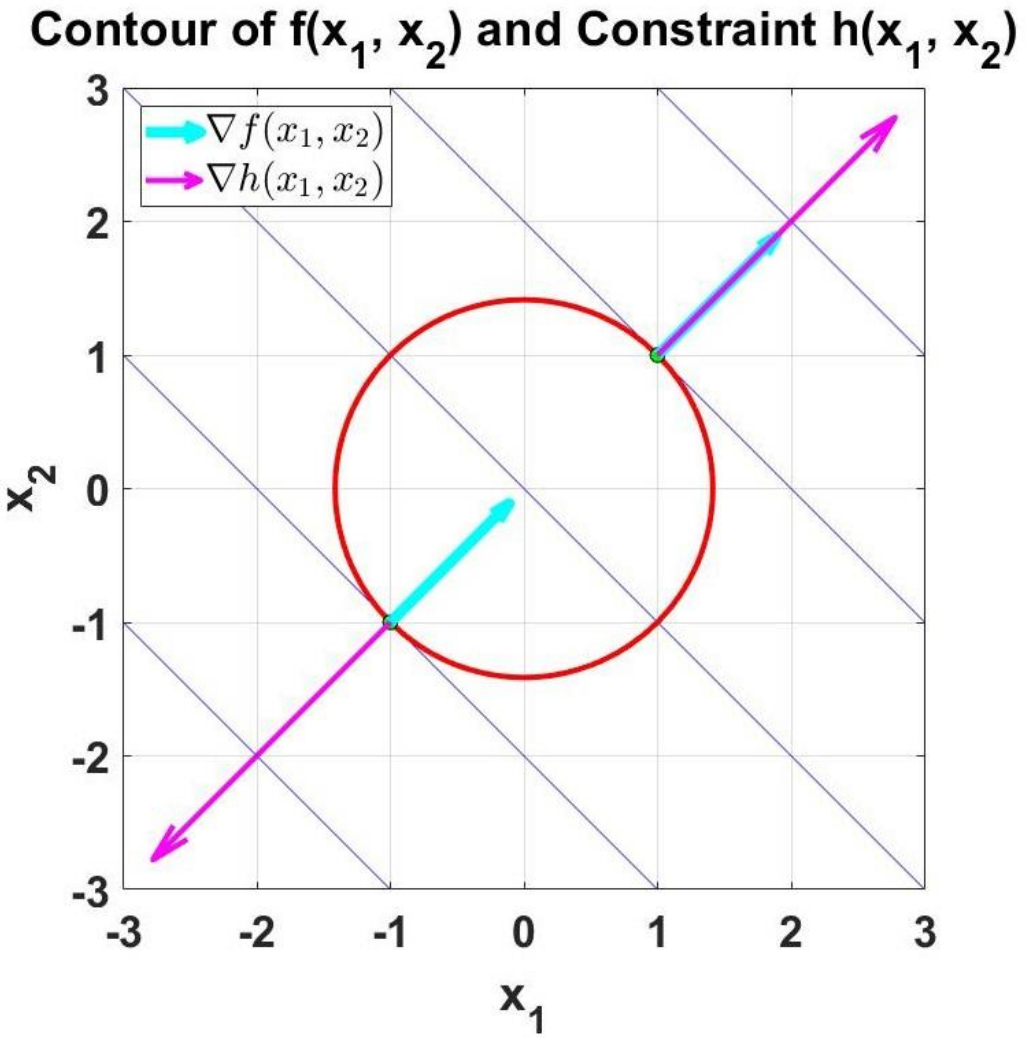
- Calculate the gradient $\nabla L(x, y, \lambda) = \begin{pmatrix} \frac{\partial L}{\partial x_1} \\ \frac{\partial L}{\partial x_2} \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x_1} + \lambda \frac{\partial h}{\partial x_1} \\ \frac{\partial f}{\partial x_2} + \lambda \frac{\partial h}{\partial x_2} \\ h \end{pmatrix} = \begin{pmatrix} 1 + 2\lambda x_1 \\ 1 + 2\lambda x_2 \\ x_1^2 + x_2^2 - 2 \end{pmatrix}$

- Now $\nabla L(x, y, \lambda) = 0 \Rightarrow \begin{cases} 1 + 2\lambda x_1 = 0 & (1) \\ 1 + 2\lambda x_2 = 0 & (2) \\ x_1^2 + x_2^2 - 2 = 0 & (3) \end{cases}$

- From (1) and (2), we have $x_1 = x_2$
- Substituting $x_1 = x_2$ in (3), we get $x_1 = x_2 = \pm 1$
- Total Lagrange points are: $(1,1), (-1,-1)$
- Function's maximum value is 2 achieved at: $(1,1)$
- Function's minimum value is -2 achieved at: $(-1,-1)$

Equality constraints: Lagrange multiplier

- Function's maximum value is 2 achieved at: $(1,1)$
- Function's minimum value is -2 achieved at: $(-1,-1)$
- By manual verification and plot helps, Extreme points are: $(1,1), (-1,-1)$



Two equality constraints:

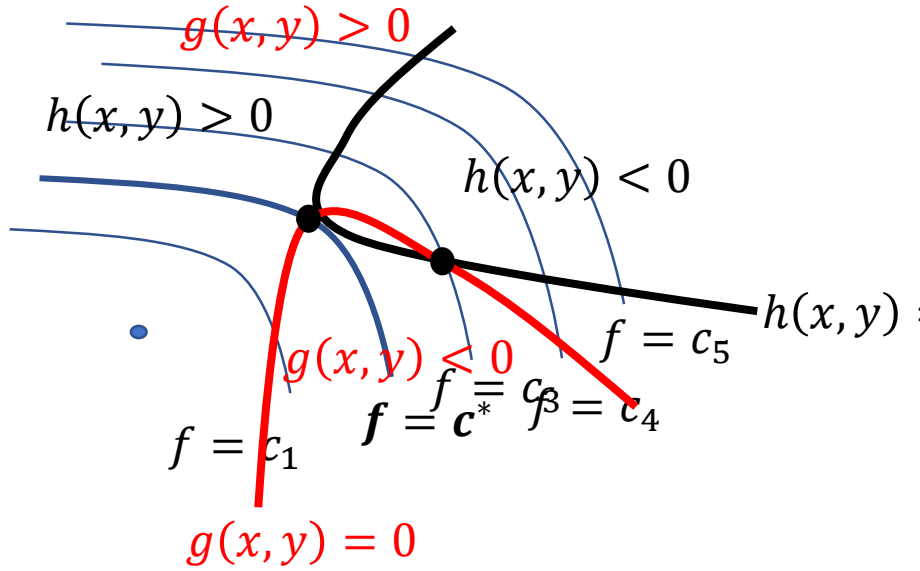
- Construct the Lagrangian function as:

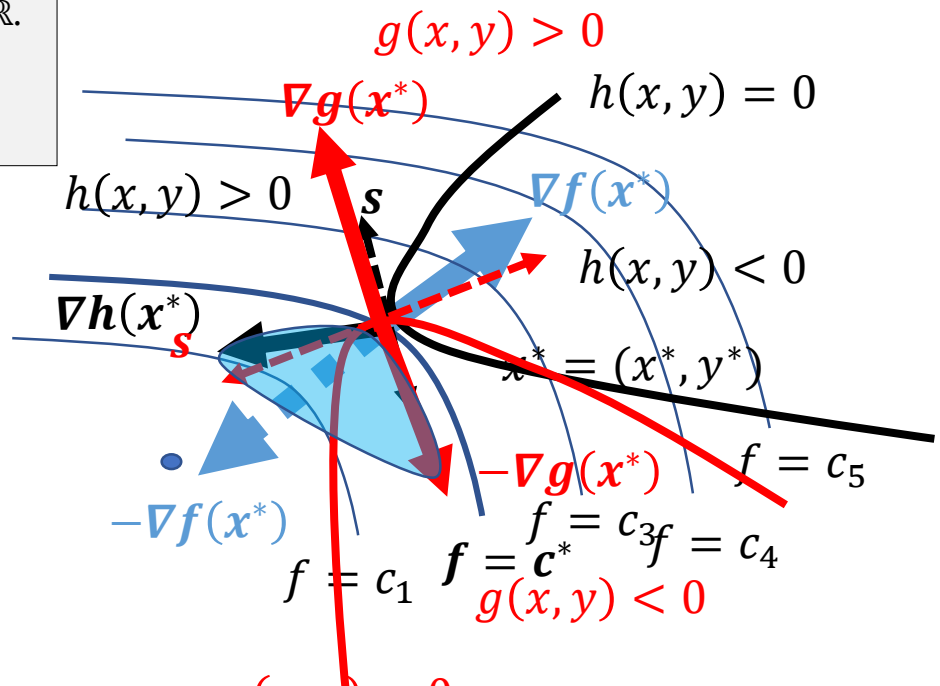
$$L(x, y, \lambda_1, \lambda_2) = f(x, y) + \lambda_1 h(x, y) + \lambda_2 g(x, y)$$

- Calculate the gradient $\nabla L(x, y, \lambda_1, \lambda_2) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda_1} \\ \frac{\partial L}{\partial \lambda_2} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda_1 \frac{\partial h}{\partial x} + \lambda_2 \frac{\partial g}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda_1 \frac{\partial h}{\partial y} + \lambda_2 \frac{\partial g}{\partial y} \\ h \\ g \end{pmatrix}$

- **Stationarity condition:**
- $\nabla_{x,y} L(x, y, \lambda_1, \lambda_2) = 0 \Rightarrow \nabla f(x, y) + \lambda_1 \nabla h(x, y) + \lambda_2 \nabla g(x, y) = 0$
- $\Rightarrow -\nabla f(x, y)$ vector is the **linear combination** of vectors $\nabla h(x, y)$ and $\nabla g(x, y)$ with $\lambda_i \in \mathbb{R}$.
- In general, $\nabla f \in span(\nabla h, \nabla g)$
- **Feasibility:** $\nabla_{\lambda_1, \lambda_2} L(x, y, \lambda_1, \lambda_2) = 0 \Leftrightarrow h(x, y) = 0, g(x, y) = 0$

For equality constraints, $\lambda_i \in \mathbb{R}$





Two equality constraints: Example

- Minimize $f(x,y) = x^2 + y^2$ under the two equality constraints:
- $h(x,y) = x + y - 1 = 0$
- $g(x,y) = x^2 + y^2 - 1 = 0$

- Construct the Lagrangian function as:

$$L(x,y,\lambda_1,\lambda_2) = f(x,y) + \lambda_1 h(x,y) + \lambda_2 g(x,y)$$
$$L(x,y,\lambda_1,\lambda_2) = x^2 + y^2 + \lambda_1(x + y - 1) + \lambda_2(x^2 + y^2 - 1\ )$$

- Calculate the gradient $\nabla L(x,y,\lambda_1,\lambda_2) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda_1} \\ \frac{\partial L}{\partial \lambda_2} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda_1 \frac{\partial h}{\partial x} + \lambda_2 \frac{\partial g}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda_1 \frac{\partial h}{\partial y} + \lambda_2 \frac{\partial g}{\partial y} \\ h \\ g \end{pmatrix} = \begin{pmatrix} 2x + \lambda_1 + 2\lambda_2 x \\ 2y + \lambda_1 + 2\lambda_2 y \\ x + y - 1 \\ x^2 + y^2 - 1 \end{pmatrix}$

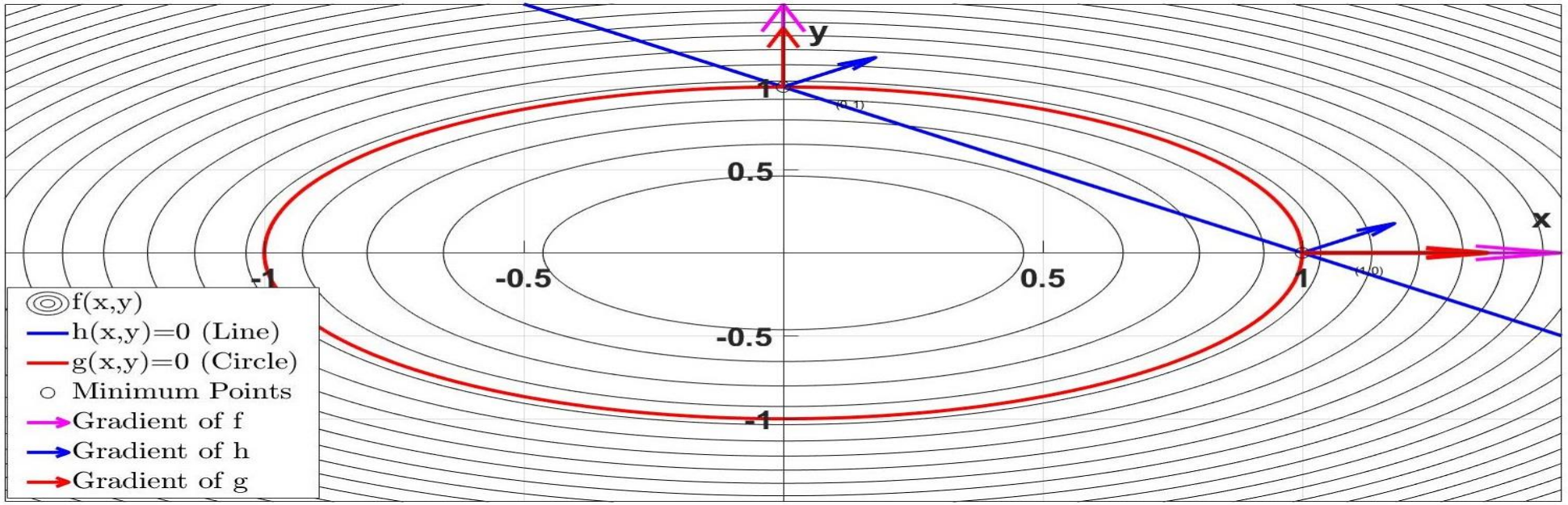
- Now $\nabla L(x,y,\lambda_1,\lambda_2) = 0 \Rightarrow \begin{cases} 2x + \lambda_1 + 2\lambda_2 x = 0 & (1) \\ 2y + \lambda_1 + 2\lambda_2 y = 0 & (2) \\ x + y - 1 = 0 & (3) \\ x^2 + y^2 - 1 = 0 & (4) \end{cases}$

Two equality constraints: Example

- From eqn (1): $2x + \lambda_1 + 2\lambda_2 x = 0 \Rightarrow \lambda_1 = -2x - 2\lambda_2 x$
  From eqn (2): $2y + \lambda_1 + 2\lambda_2 y = 0 \Rightarrow \lambda_1 = -2y - 2\lambda_2 y$
  $\Rightarrow x(1 + \lambda_2) = y(1 + \lambda_2)$
  - Case 1: If $(1 + \lambda_2) \neq 0$ then $x = y$
    - From (3), $x + y - 1 = 0 \Rightarrow 2x - 1 = 0 \Rightarrow x = \frac{1}{2}, y = \frac{1}{2}$
    - From (4), $x^2 + y^2 - 1 = 0 \Rightarrow \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 - 1 \neq 0$ (i.e., $x = \frac{1}{2}, y = \frac{1}{2}$ do not satisfy second constraint)
    - It means this case is not possible.
  - Case 2: If $(1 + \lambda_2) = 0 \Rightarrow \lambda_2 = -1$ (-1 weight for $g(x, y) = 0$)
    - Substitute $\lambda_2 = -1$ in eqn (1), we get: $2x + \lambda_1 - 2x = 0 \Rightarrow \lambda_1 = 0$ (zero weight for $h(x, y) = 0$)
    - From (3): $x + y - 1 = 0 \Rightarrow y = 1 - x$. Put this value in eqn (4) we get: $x^2 + (1 - x)^2 = 1 \Rightarrow x = 0,1$
    - $\Rightarrow y = 1,0$
    - Thus, we have two ***candidate optimal points*** (or stationary points or Lagrange points): $(x, y) = (0,1)$ and $(x, y) = (1,0)$

- By manual verification and plot helps:
- At $(0,1)$: $f(0,1) = 0^2 + 1^2 = 1$
- At $(1,0)$: $f(1,0) = 1^2 + 0^2 = 1$
- Both points yield $f(x, y) = 1$, so the minimum value of $f(x, y)$ subject to the constraints is 1, achieved at both $(0,1)$ and $(1,0)$.

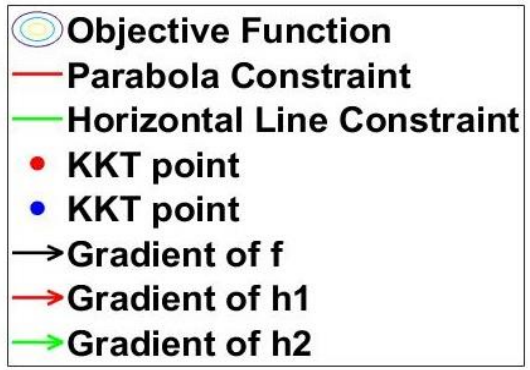
Two equality constraints: Example

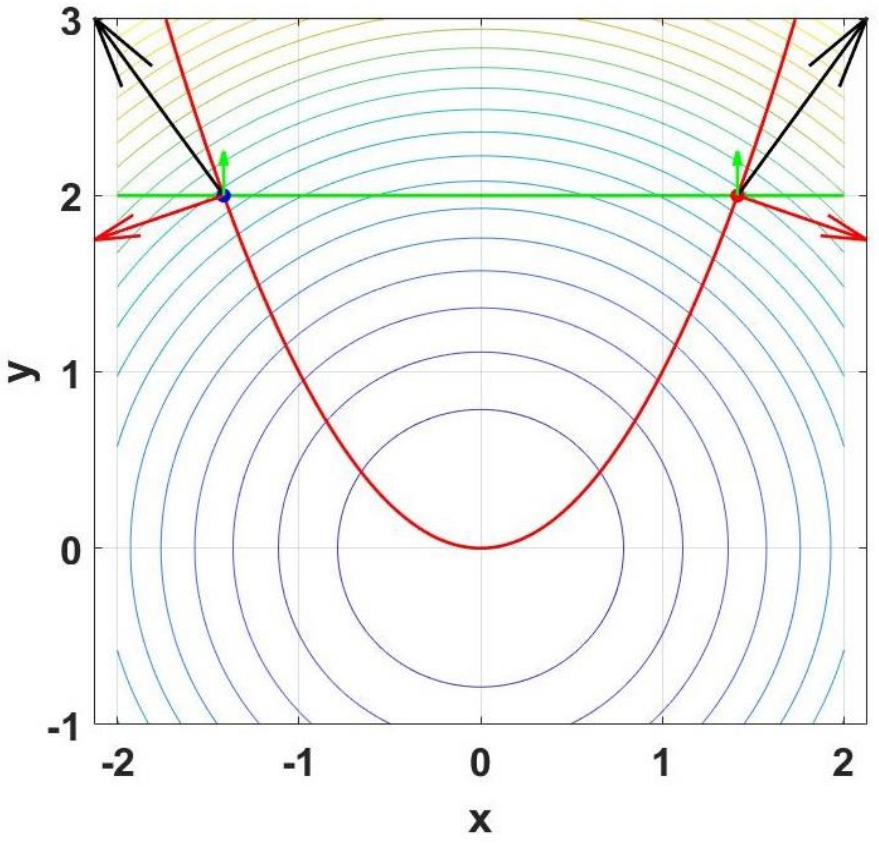


- $\nabla f = (2x, 2y)^T, \nabla h = (1,1)^T, \nabla g = (2x, 2y)^T, \lambda_1 = 0, \lambda_2 = -1$
- At point $(1,0)$
  - $\nabla f = (2,0)^T, \nabla h = (1,1)^T, \nabla g = (2,0)^T \Rightarrow -(2,0)^T = 0 \times (1,1)^T - 1(2,0)^T$
  - $-\nabla f(x,y) = \lambda_1 \nabla h(x,y) + \lambda_2 \nabla g(x,y)$ satisfied.
- At point $(0,1)$
  - $\nabla f = (0,2)^T, \nabla h = (1,1)^T, \nabla g = (0,2)^T \Rightarrow -(0,2)^T = 0 \times (1,1)^T - 1(0,2)^T$
  - $-\nabla f(x,y) = \lambda_1 \nabla h(x,y) + \lambda_2 \nabla g(x,y)$ satisfied.

Two equality constraints: DIY

- Minimize $f(x,y) = x^2 + y^2$ under the two equality constraints:
- $h(x,y) = y - x^2$
- $g(x,y) = y - 2$
- Use the method of Lagrange multipliers to find the Lagrange points (x,y) and the corresponding multipliers (λ_1, λ_2) .





- First Lagrange point is $(\sqrt{2}, 2)$ with Lagrange multipliers $(\lambda_1, \lambda_2) = (1, -3)$
- Second points point is $(-\sqrt{2}, 2)$ with Lagrange multipliers $(\lambda_1, \lambda_2) = (-1, -1)$

Example 1:

- Minimize $f(x,y) = x^2 + y^2$ under the constraint:
- $g(x,y) = x^2 + y^2 - 1$
- Use the method of Lagrange multipliers to find the Lagrange points (x,y) and the corresponding multiplier λ.

- Construct the Lagrangian function as:

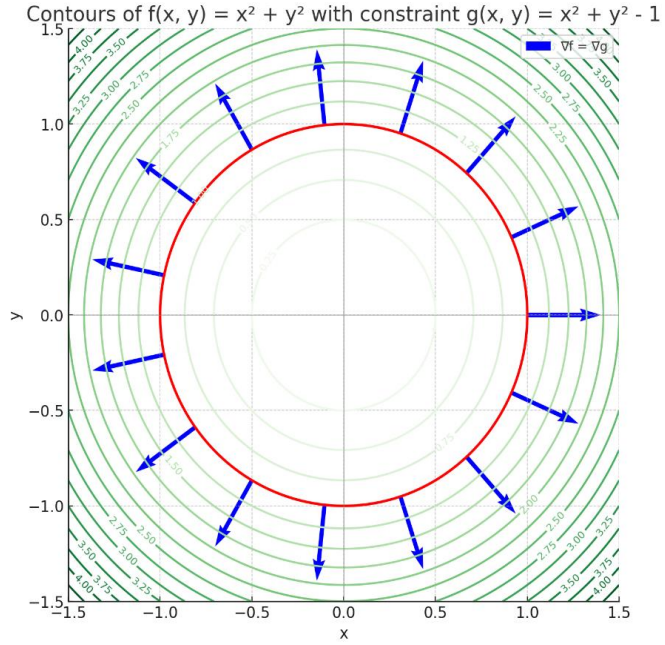
$$L(x,y,\lambda) = f(x,y) + \lambda g(x,y)$$
$$L(x,y,\lambda) = x^2 + y^2 + \lambda(x^2 + y^2 - 1\ )$$

- Calculate the gradient $\nabla L(x,y,\lambda) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda \frac{\partial g}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda \frac{\partial g}{\partial y} \\ g \end{pmatrix} = \begin{pmatrix} 2x + 2\lambda x \\ 2y + 2\lambda y \\ x^2 + y^2 - 1 \end{pmatrix}$

- Now $\nabla L(x,y,\lambda) = 0 \Rightarrow \begin{cases} 2x + 2\lambda x = 0 & (1) \\ 2y + 2\lambda y = 0 & (2) \\ x^2 + y^2 - 1 = 0 & (3) \end{cases}$

Example 1: (contd.)

- Now $\nabla L(x, y, \lambda) = 0 \Rightarrow \begin{cases} 2x + 2\lambda x = 0 & (1) \\ 2y + 2\lambda y = 0 & (2) \\ x^2 + y^2 - 1 = 0 & (3) \end{cases}$
- From equ (1) and (2) $\Rightarrow \begin{cases} x(1+\lambda) = 0 & (4) \\ y(1+\lambda) = 0 & (5) \end{cases}$
  - Case 1: $(1+\lambda) \neq 0 \Rightarrow x = 0$ and $y = 0 \Rightarrow$ But this contradicts the constraint $x^2 + y^2 - 1 = 0$
  - Case 2: $(1+\lambda) = 0 \Rightarrow \lambda = -1$
    - From equ (1) and (2) $\Rightarrow \begin{cases} 2x - 2x = 0 \Rightarrow x(2-2) = 0 & (6) \\ 2y - 2y = 0 \Rightarrow y(2-2) = 0 & (7) \end{cases}$
      - Equ (6): always satisfied by any x
      - Equ (7): always satisfied by any y
        - It means stationarity conditions are trivially satisfied for all values of x, y.
    - The constraint condition $x^2 + y^2 - 1 = 0$ still holds.
  - When $\lambda = -1$, the Lagrange condition is satisfied for any point on the unit circle $x^2 + y^2 - 1 = 0$. The Lagrange points are like (1,0), (0,1) and so on. Every feasible point is a Lagrange point here.
  - All Lagrange points are optimal in this case (manual verification and graph helps) .



Example 2: Lagrange Points and Constraint Qualification

- Minimize $f(x,y) = x + y$ under the constraint:
- $h(x,y) = x^2 + y^2 = 0$

- Construct the Lagrangian function as:

$$L(x,y,\lambda) = f(x,y) + \lambda h(x,y)$$
$$L(x,y,\lambda) = x + y + \lambda(x^2 + y^2)$$

- Calculate the gradient $\nabla L(x,y,\lambda) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} + \lambda \frac{\partial h}{\partial x} \\ \frac{\partial f}{\partial y} + \lambda \frac{\partial h}{\partial y} \\ h \end{pmatrix} = \begin{pmatrix} 1 + 2\lambda x \\ 1 + 2\lambda y \\ x^2 + y^2 \end{pmatrix}$

- Now $L(x,y,\lambda) = 0 \Rightarrow \begin{cases} 1 + 2\lambda x = 0 & (1) \\ 1 + 2\lambda y = 0 & (2) \\ x^2 + y^2 = 0 & (3) \end{cases}$

Example 2 contd.

- Since $x^2 \geq 0, y^2 \geq 0$
- Using eqn (3): $x^2 + y^2 = 0 \Rightarrow x = 0, y = 0$
- Substitute into equations (1) and (2).
  - Eqn(1) becomes:
    - $1 + 2\lambda(0) = 0 \Rightarrow 1 = 0$
  - Eqn(2) becomes:
    - $1 + 2\lambda(0) = 0 \Rightarrow 1 = 0$
- Both equations lead to a contradiction.
- Step 3: Conclusion:
- The constrained optimization problem has a global minimum at (0,0), since it is the only feasible point.
- However, no Lagrange multiplier λ satisfies the stationarity conditions at this point.
- Since the **constraint qualification (CQ)** fails here!!!

Example 2 contd.

- Why constraint qualification (CQ) fails here?
- $\nabla h(0,0) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}_{(0,0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- $\nabla f(0,0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{(0,0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $\nabla f(0,0) + \lambda \nabla h(0,0) = 0 \Rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, which is impossible for every value of λ, including $\lambda = 0$
- Therefore, the Lagrange multiplier condition is not satisfied, even though a global minimum exists, because the constraint qualification (CQ) fails. so (0,0) is not a Lagrange point.

Key Takeaways: Lagrange Points and Constraint Qualification

- Lagrange points are not necessarily local optima. They may correspond to local minima, local maxima, or saddle points.
- **With CQ:** If a local optimum satisfies an appropriate CQ, then it **must be (guaranteed)** a Lagrange point.
  - **Local optimum + CQ → Lagrange point**
- **Without CQ:** A local optimum **may or may not be** a Lagrange point. The existence of a local optimum does not guarantee that the Lagrange multiplier stationarity conditions can be satisfied.
  - **Local optimum + CQ fails → Lagrange point may or may not exist**
- Moving forward: The Karush–Kuhn–Tucker (KKT) conditions generalize the Lagrange multiplier approach to problems **involving inequality constraints as well as equality constraints.**

Common CQ

- In general, regularity means that the constraints are **well-behaved at a feasible point i.e.,** they are not redundant, conflicting, or degenerate. Most common regularity conditions are:
  - **Linear Independence Constraint Qualification (LICQ)**
  - **Slater condition (for convex problems)**
  - Mangasarian-Fromovitz Constraint Qualification (MFCQ)
  - Constant Rank Constraint Qualification (CRCQ)

CQ	Applies To	Condition	Purpose	Strength
LICQ	General problems	Gradients of active constraints are **linearly independent**	Ensures KKT + unique multipliers	Strongest
MFCQ	General problems	Gradients of Equality constraints ∇h_i independent, and ∃ direction that **strictly decreases** all active g_j	Ensures KKT holds	Weaker than LICQ
Slater	Convex problems only	∃ strictly feasible point:$g_j(x) < 0$	Guarantees KKT + strong duality	Simple check
CRCQ	General problems	Active constraint gradients maintain **constant rank nearby**	General KKT applicability	Most general

Constraint qualification: LICQ

Minimize $f(\boldsymbol{x})$

Subject to:

$$h_j(x) = 0, \quad j = 1,2,..,l$$

- At a feasible point x^*, the LICQ **(Linear Independence Constraint Qualification)** condition requires that the rows of the Jacobian matrix $J_h(x^*)$of the **active constraints** are linearly independent.

$$J_h(x^*) = \begin{bmatrix} \nabla h_1(x^*) \\ \nabla h_2(x^*) \\ \nabla h_3(x^*) \\ \\ \\ \nabla h_l(x^*) \end{bmatrix}$$

- where $\nabla h_i(x^*)$ is the gradient (row vector) of the ith constraint with respect to x at x^*.
- Or Rank $\left(J_h(x^*)\right) = l$ or At a feasible point x^*, LICQ holds if the rows of $J_h(x^*)$ (i.e., $\nabla h_1(x^*)$, $\nabla h_2(x^*)$, $\nabla h_3(x^*)$,…, $\nabla h_l(x^*)$ are linearly independent.

Minimize $f(\boldsymbol{x_1}, \boldsymbol{x_2}) = \boldsymbol{x_1} + \boldsymbol{x_2}$

Subject to:

$$h_1(x, y) = x_1 - 1 = 0$$
$$h_2(x, y) = x_2 - 1 = 0$$

- $J_h(x, y) = \begin{bmatrix} \nabla h_1(x, y) \\ \nabla h_3(x, y) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- Rank $J_h(x, y) = 2$ (no of active constraints), LICQ holds.

Minimize $f(x_1, x_2) = x_1 + x_2$

Subject to:

$$h_1(x, y) = x_1 + x_2 - 1 = 0$$
$$h_2(x, y) = 2x_1 + 2x_2 - 2 = 0$$

$$J_h(x, y) = \begin{bmatrix} \nabla h_1(x, y) \\ \nabla h_3(x, y) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

- Rank $J_h(x, y) = 1 \neq 2$ (no of active constraints), LICQ do not hold here.

Without CQ: case 1

- Minimize $f(x,y) = 2x + 3y$ under the equality constraints:
- $h(x,y) = x^2 + y^2 = 0$

- Construct the Lagrangian function as:

$$L(x,y,\lambda_1) = f(x,y) + \lambda_1 h(x,y)$$
$$L(x,y,\lambda_1) = 2x + 3y + \lambda_1(x^2+y^2)$$

- Calculate the gradient $\nabla L(x,y,\lambda_1) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda_1} \end{pmatrix} = \begin{pmatrix} 2 + 2\lambda_1 x \\ 3 + 2\lambda_1 y \\ h \end{pmatrix}$
- Now $\nabla L(x,y,\lambda_1) = 0 \Rightarrow \begin{cases} 2 + 2\lambda_1 x = 0 & (1) \\ 3 + 2\lambda_1 y = 0 & (2) \\ x^2 + y^2 = 0 & (3) \end{cases}$

- From eqn (3), $x^2 + y^2 = 0$
  - The only solution to this equation is $x = 0$ and $y = 0$, as both squared terms must be zero to satisfy the equality. The feasible set is reduced to a single point $(x,y) = (0,0)$.
- Now put $(x,y) = (0,0)$ in eqn (1) and (2):
  - $2 = 0$ and $3 = 0$ contradiction. Or we can not find any λ
- **This contradiction tells us that the Lagrange condition fail to apply here, and we can't derive a meaningful solution using them.**

- Here $(x,y) = (0,0)$, $\nabla h(x,y) = [0,0]^T$, $\nabla f(x,y) = [2,3]^T$
- The stationarity condition: $\nabla f(x,y) + \lambda_1 \nabla h(x,y) = 0$ does not depend on λ at all. No value of λ can satisfy this condition meaningfully because the $\nabla h(x,y) = [0,0]^T$, meaning the feasible direction for optimization is zero as well. It means no valid (non-zero) gradient to describe the constraint direction.
- **Thus, while (0,0) is indeed the optimum for the problem, it does not satisfy the Lagrange conditions, showing that KKT is not always applicable, particularly in cases where constraint qualifications (like LICQ) are violated**. Or **optimal points are not necessarily KKT points when regularity conditions are violated**.

zero vector is never linearly independent

Constraint qualifications (like LICQ)?.

Without CQ: case 2

- Minimize $f(x,y) = x^2$ under the equality constraints:
- $h(x,y) = y^2 = 0$

- Construct the Lagrangian function as:

$$L(x,y,\lambda_1) = f(x,y) + \lambda_1 h(x,y)$$
$$L(x,y,\lambda_1) = x^2 + \lambda_1 y^2$$

- Calculate the gradient $\nabla L(x,y,\lambda_1) = \begin{pmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial y} \\ \frac{\partial L}{\partial \lambda_1} \end{pmatrix} = \begin{pmatrix} 2x \\ 2\lambda_1 y \\ y^2 \end{pmatrix}$
- Now $\nabla L(x,y,\lambda_1) = 0 \Rightarrow \begin{cases} 2x = 0 & (1) \\ 2\lambda_1 y = 0 & (2) \\ y^2 = 0 & (3) \end{cases}$

- From eqn (3), $y^2 = 0 \Rightarrow y = 0$
- From eqn (1), $2x = 0 \Rightarrow x = 0$
- From eqn (2), $2\lambda_1 y = 0 \Rightarrow 2\lambda_1(0) = 0$
  - which is true for any λ_1. So λ_1 is **not determined** (multipliers are non-unique).
  - So the stationary (Lagrange/KKT) points are $(x,y) = (0,0)$ with λ_1 arbitrary.

- The feasible set $\{(x,0): x \in \mathbb{R}\}$, since $y^2 = 0 \Leftrightarrow y = 0$
- On the feasible set $\{(x,0): x \in \mathbb{R}\}$, $f(x,0) = x^2$, which attains its minimum at $x = 0$. Thus (0,0) is a (indeed global in this case) minimizer.
- At (0,0):
  - $\nabla f(x,y) = (2x,0)^T = (0,0)^T$
  - $\nabla h(x,y) = (0,2y)^T = (0,0)^T$ CQ fails here. (A local optimum **may or may not be** a Lagrange/KKT point, i.e., guarantee lost)
  - However, here $\nabla f(0,0)^T = -\lambda_1 \nabla h(0,0)^T$, for any λ.
  - CQ fails, but stationarity can still hold (accidentally), often with non-unique multipliers.

Problem with equality and **inequality** constraints

(Generalization of Lagrange multipliers)

Inequality constraints: Active constraints

Minimize $f(\boldsymbol{x})$

subject to:

$$g_i(x) \leq 0, \quad i = 1,2,..,m$$

Where $x \in \Omega \subseteq \mathbb{R}^n$ and f, g_i, usually assumed to be in C^2 (i.e., with continuous second partial derivatives)

Active constraint:

- An inequality constraint $g_i(x) \leq 0$ is active (binding) **at a point x^*** if: $g_i(x^*) = 0$, otherwise it is inactive (not binding) constraint.
- Active constraint either touches or interacts with the level curves of the objective function at the optimal point.
- Equality constraints are always active at any feasible point.
- The active set $A(x^*)$ at a point x^* is defined as:

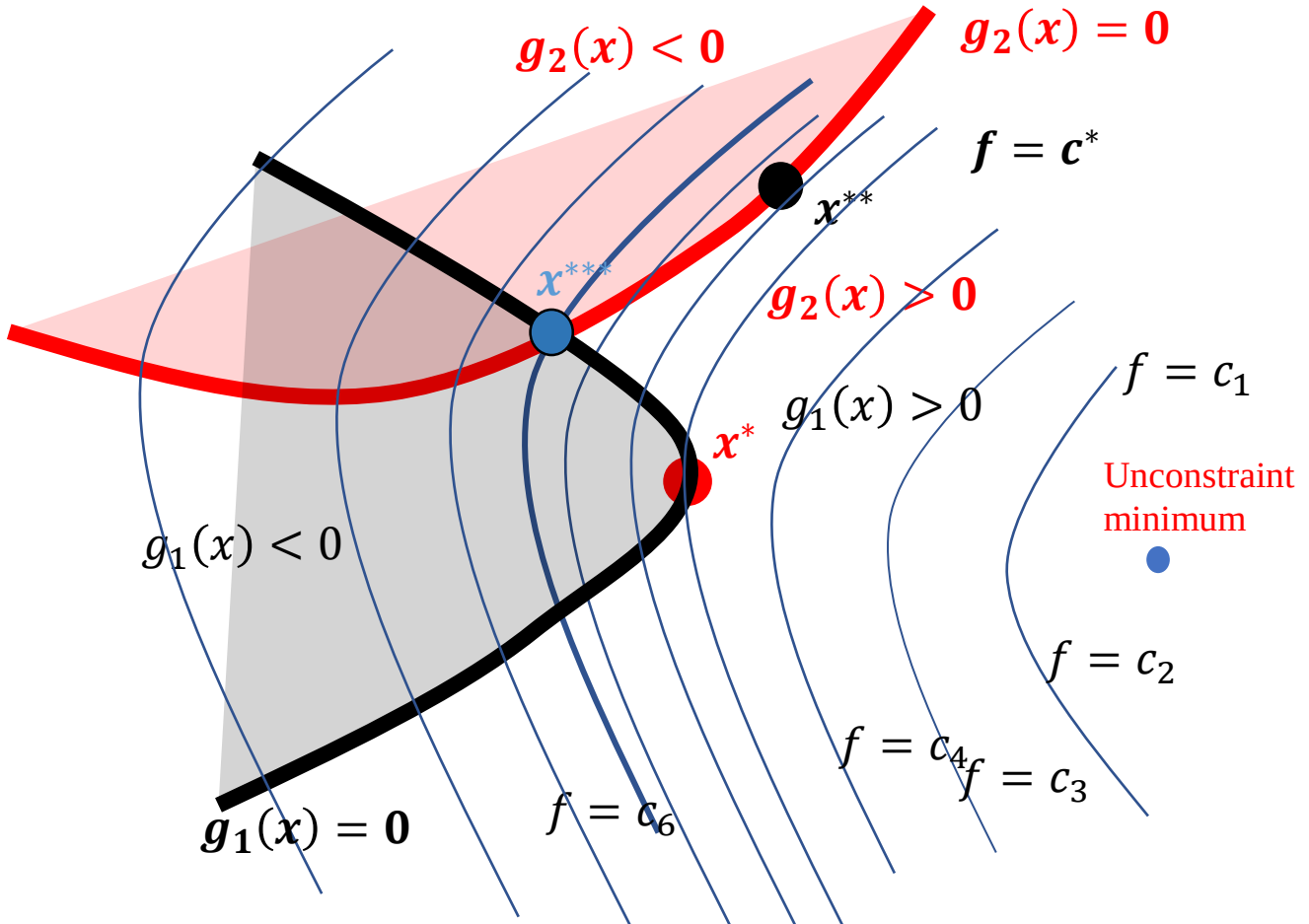
$$A(x^*) = \{j: h_j(x^*) = 0\} \cup \{i: g_i(x^*) = 0\}$$

Minimize $f(\boldsymbol{x})$

subject to:

$$g_1(x) \leq 0$$
$$g_2(x) \leq 0$$

- At x^*: $g_1(x)$ is active, $g_2(x)$ is not active
- At x^{**}: $g_2(x)$ is active, $g_1(x)$ is not active
- At x^{***}: Both $g_1(x)$ and $g_2(x)$ are active

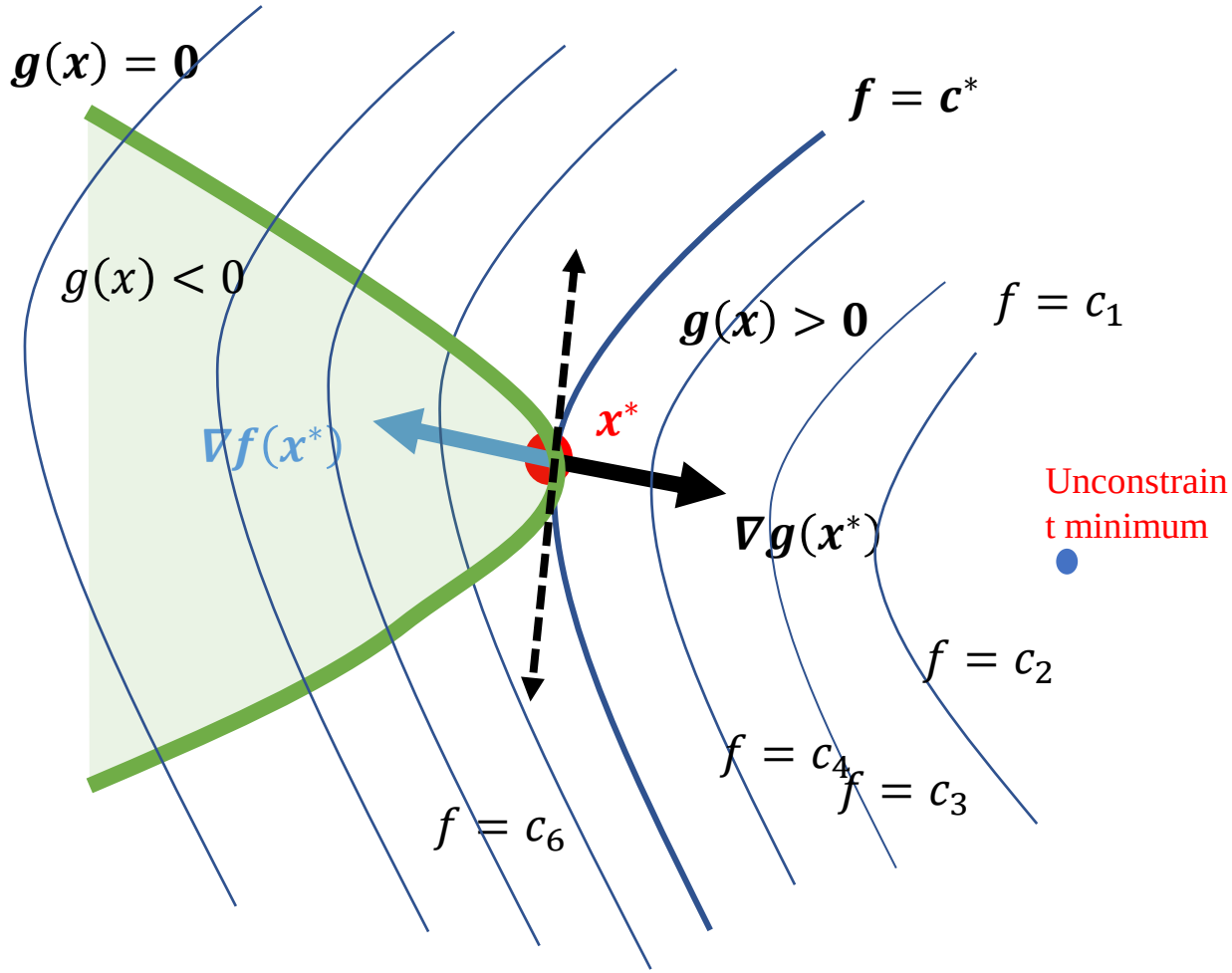


Inequality constraints:

Minimize $f(\boldsymbol{x})$
subject to:
$g(x) \leq 0$

Case 1: When the unconstraint minimum x^* is outside the feasible region

- At constraint optima x^*, the contour $g(x) = 0$ and function level curve $f = c^*$ are tangential.
- At x^*, $g(x^*) = 0$: constraint $g(x)$ is active. It means, $g(x)$ must have a **positive influence** ($\lambda^* > 0$) on the optimal solution. Or $\lambda^* \, g(x^*) = 0$ for some $\lambda^* > 0$.
- In the whole feasible region ($g(x) \leq 0$), moving away from x^*, the functional value increases.



At constraint optima x^*, $\nabla f(x^*)$ and $\nabla g(x^*)$ must be in **opposite direction** (unlike the equality constraint). It means: (stationarity condition)

$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0$ for some $\lambda^* > 0$

Inequality constraints:

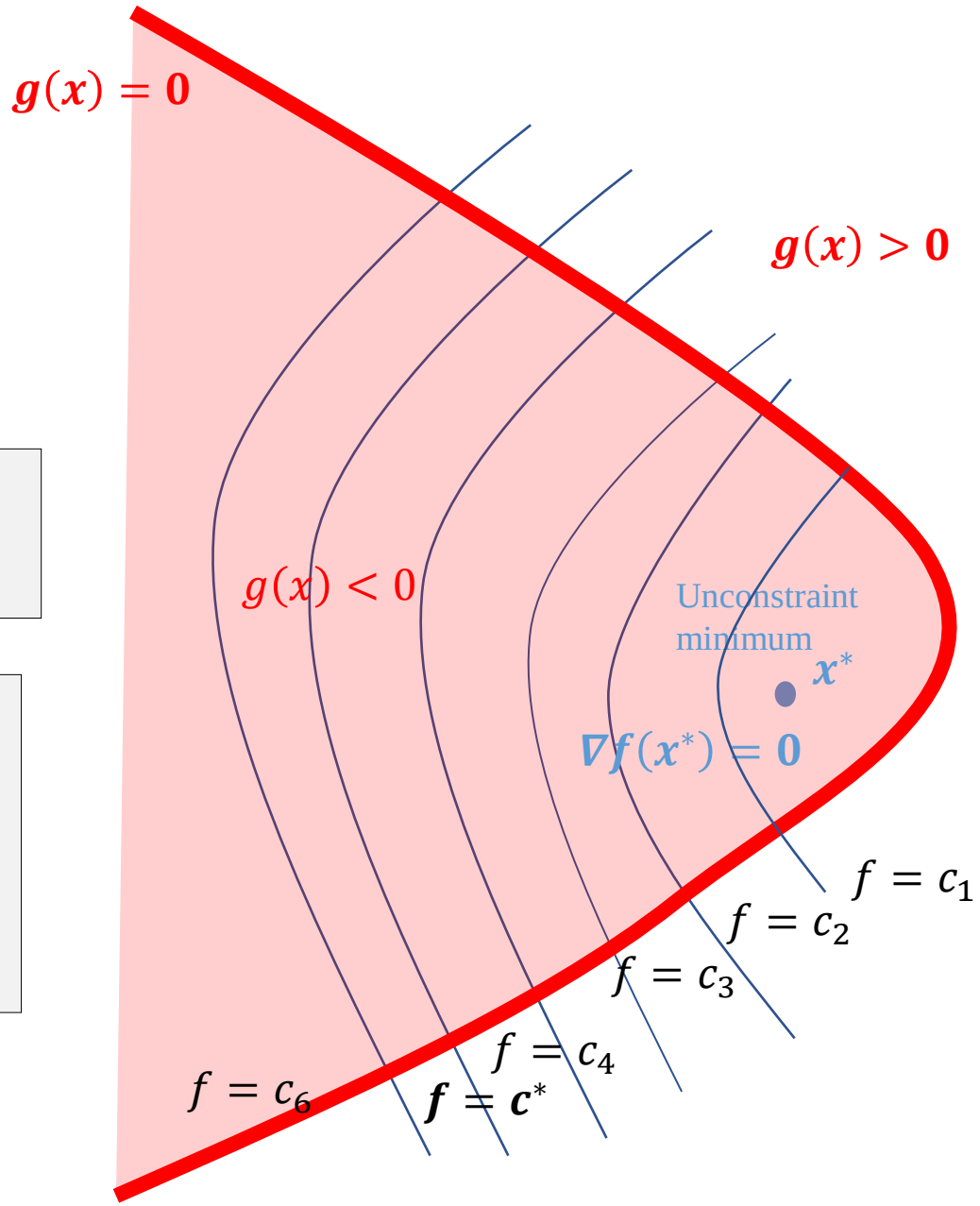
Minimize $f(\boldsymbol{x})$
subject to:
$g(x) \leq 0$

Case 2: When the unconstraint minimum x^* is inside the feasible region

The constraint $g(x)$ is inactive since $g(x^*) \neq 0$
It means $\lambda^* g(x^*) = 0 \Longrightarrow \lambda^* = 0$ ($g(x)$ must have a **zero influence** ($\lambda^* = 0$) on the optimal solution)

- The unconstraint minimum of f is also a constraint minimum x^*.
- x^* is unconstraint minimum of f means $\nabla f(x^*) = 0$
- x^* is constraint minimum of f satisfies: (stationarity condition)

$$\nabla f(x^*) + \lambda^* \nabla g(x^*) = 0 \Longrightarrow \lambda^* = 0$$



Two inequality constraints:

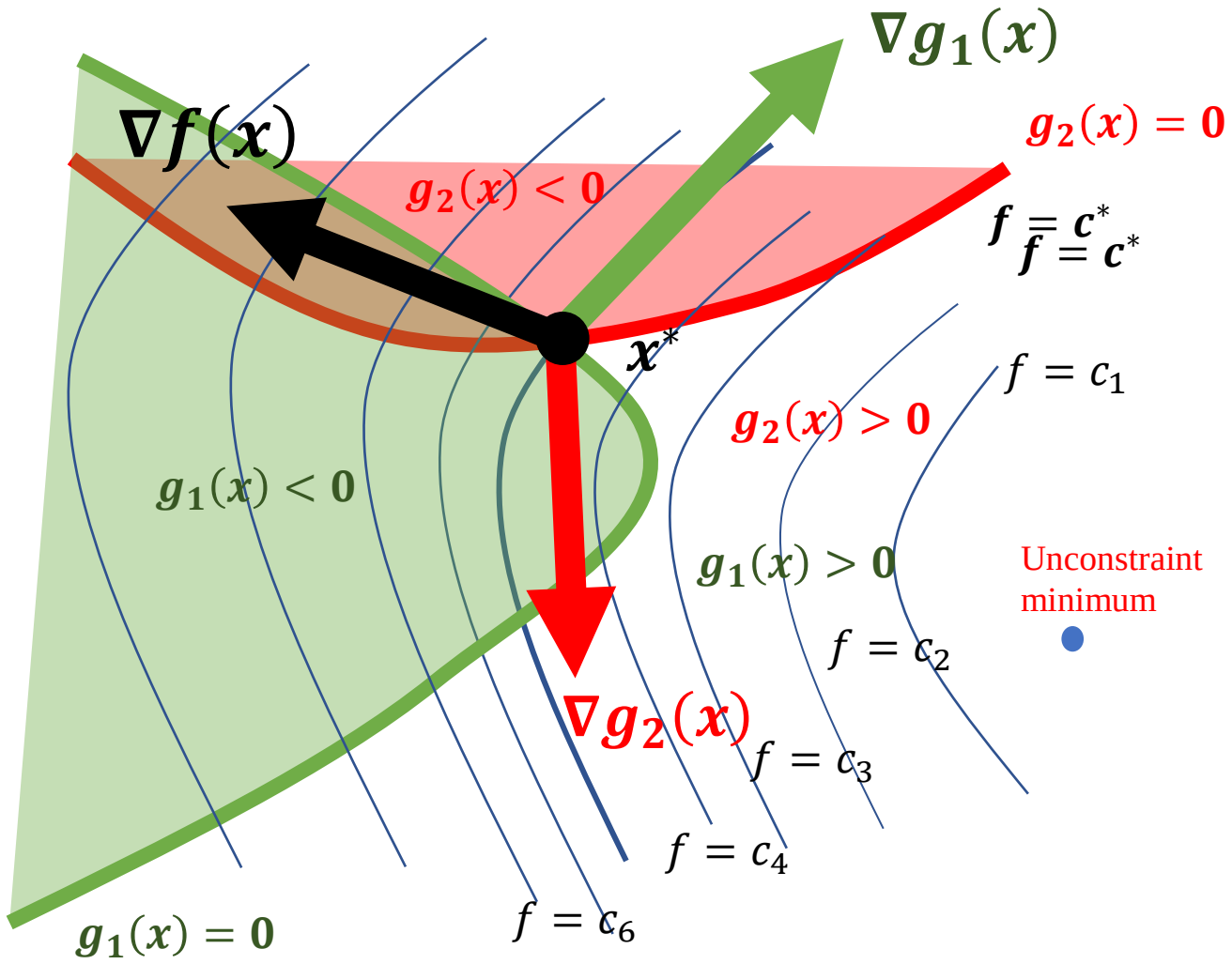
Minimize $f(\boldsymbol{x})$
subject to:
$g_1(x) \leq 0$
$g_2(x) \leq 0$

Feasible region: $\mathcal{F} = \{x \in \mathbb{R}^n | g_1(x) \leq 0 \text{ and } g_1(x) \leq 0\}$

- **At constraint minimum** $\boldsymbol{x}^*$ of given function f:
- $\nabla f(x^*) + \lambda_1^* \nabla g_1(x^*) + \lambda_2^* \nabla g_2(x^*) = 0$ with
  - $\lambda_1^*, \lambda_2^* > 0$.
  - Both $g_1(x^*)$ and $g_2(x^*)$ are **active** ($g_1(x^*) = g_2(x^*) = 0$)
- $\nabla f(x^*)$ is a linear combination of the gradients of the **active** constraints ($g_i(x^*) = 0$)

Angles between $\nabla f(x^*)$, and gradients of active constraints $\nabla g_1(x^*)$ and $\nabla g_2(x^*)$ should be obtuse (more than 90) for **being $\boldsymbol{x}^*$ as a constraint minimum**.

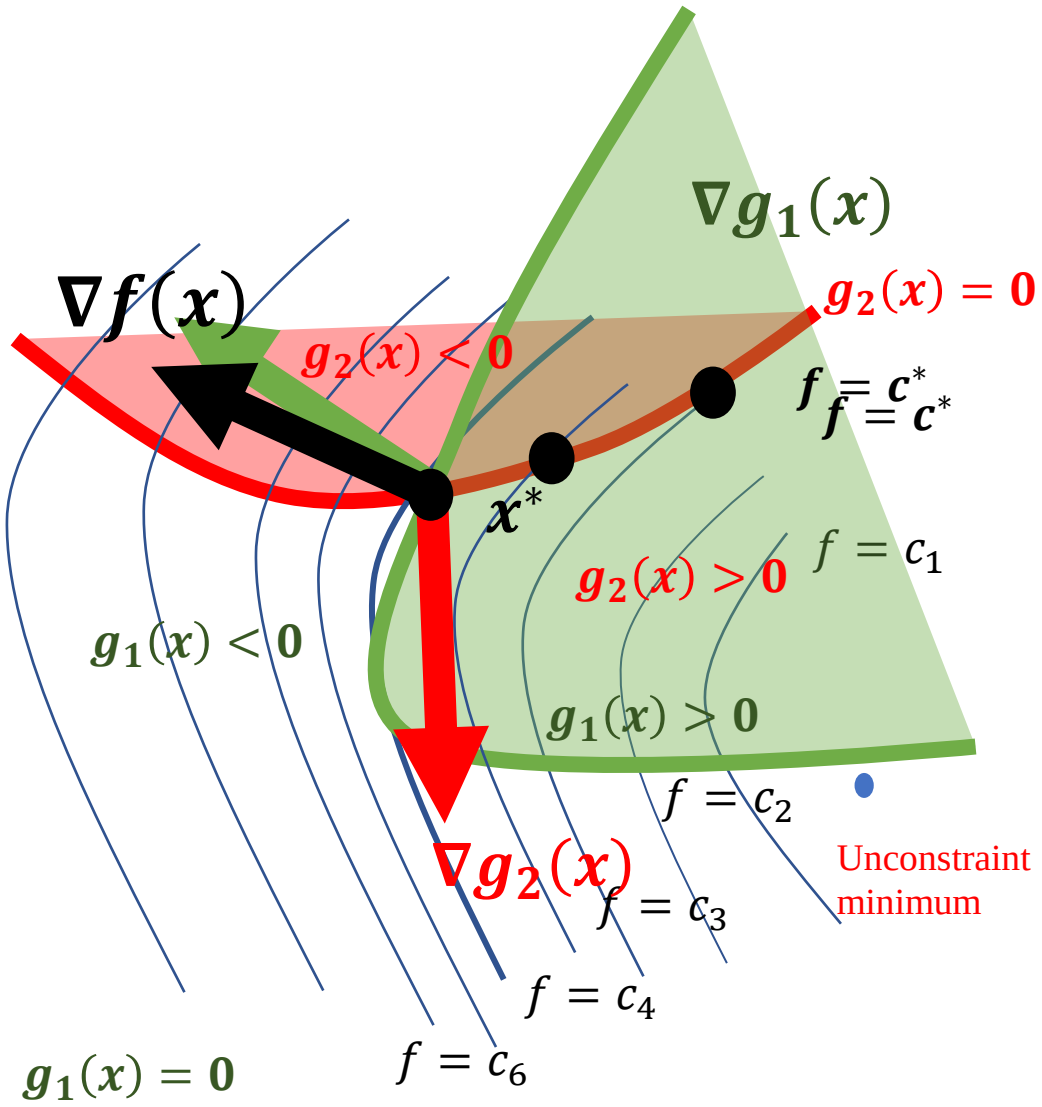
- $\nabla f(x^*)^T \nabla g_i(x^*) < 0, \forall$ active constraint i



Two inequality constraints:

- For x^* **to be a constraint minimum:**
  - $\nabla f(x^*)^T \nabla g_i(x^*) < 0, \forall$ active constraint i
- It ensures that moving along the feasible boundary defined by $g_i(x^*) = 0$ will not decrease $f(x)$ further.
- If the angle were acute ($\nabla f(x^*)^T \nabla g_i(x^*) > 0$), there would be a direction within the feasible region where $f$could still decrease, meaning x^* wouldn't be a minimum.

- x^* must lie on the **boundary of the feasible region** where at least one constraint $g_i(x) \leq 0$ is active (i.e., $g_i(x^*) \leq 0$).
- If x^* is strictly inside the feasible region, constraint gradients $\nabla g_i(x^*)$ don't affect the solution. In the case, x^* would satisfy **unconstrained optimality**, requiring $\nabla f(x^*) = 0$.
- For a constrained minimum, the stationarity condition link $\nabla f(x^*)$ with the gradients of **active constraints** at x^*.
- A true constrained minimum occurs at the edge of the feasible region, where constraint gradients influence the solution; otherwise, it behaves as an unconstrained minimum.



KKT (*Karush-Kuhn-Tucker*) conditions

(First order optimality conditions or necessary conditions)

- For a constrained optimization problem

$$\text{Minimize } f(\boldsymbol{x})$$

subject to:

$$g_i(x) \leq 0, \quad i = 1,2,..,m$$
$$h_j(x) = 0, \quad j = 1,2,..,l$$

Where $x \in \Omega \subseteq \mathbb{R}^n$ and f, g_i, h_i usually assumed to be in C^2 (i.e., with continuous second partial derivatives)

- For a point or solution x^* to be a local minimizer, it must satisfy certain conditions:

- **Necessary conditions: (KKT conditions):**
- For x^* to be a **local** minimizer, it must satisfy the KKT conditions (assuming a constraint qualification, like LICQ, holds):
  - **Stationarity:**
  - Gradient of the Lagrangian with respect to x^* must be zero:
  - The gradient of f balances with the gradient of the constraints:

$$\nabla f(x^*) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^{l} \mu_j \nabla h_j(x^*) = 0$$

    - Where λ_i are multipliers for inequality constraints ($\lambda_i \geq 0$), and $\mu_j \in \mathbb{R}$ are multipliers for equality constraints
    - ***The optimal point x^* cannot be improved without violating a constraint. The gradient of the objective is "balanced" by the gradients of the constraints , hence, no feasible descent direction exists,***
  - **Primal Feasibility: x^* satisfies the constraints:**

$$\boldsymbol{g_i(x^*) \leq 0, \forall i}$$
$$\boldsymbol{h_j(x^*) = 0, \forall j}$$

KKT (*Karush-Kuhn-Tucker*) conditions

(First order optimality conditions or necessary conditions)

- **Dual feasibility:**

$$\lambda_i \geq 0, i = 1,2, \dots, m$$

(Non-negative Lagrange multipliers for inequality constraints)

- **Complementary slackness:**

$$\lambda_i g_i(x^*) = 0, i = 1,2,3, \dots, m$$

(i.e., either a inequality constraint is active or its multiplier is zero).

- These conditions are **necessary**, so any local minimizer x^* **must** follow them. But satisfying KKT doesn't guarantee x^* is the minimizer. It could be a **maximizer or saddle point.**